An Empirical Study of Aggregation Methods Under Verifier Capture in Multi-LLM Ensembles: When Does Majority Voting Fail, and Can We Beat It?

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Abstract

Multi-LLM verification ensembles increasingly aggregate the verdicts of several LLM *verifiers* to reduce variance and adversarial brittleness relative to any single model. A growing concern in such ensembles is *verifier capture*: a small coalition of verifiers whose outputs are systematically biased can flip the aggregated decision while remaining individually plausible. This paper contributes (i) a formal definition of capture-resilient voting (CRV) and the threat model distinguishing prompt-injection, rubric-injection, and majority-bias mimicry attacks; (ii) an experimental protocol comparing seven aggregation methods (plain majority, PBFT-style deterministic thresholding, uniform-weighted mean, three trust-score variants, and an adaptive hybrid) across verifier pools of size $N \in \{5, 7, 10\}$ with capture counts $f \in \{1, 2, 3\}$ in a synthetic benchmark and a small local-model pilot; and (iii) a working hypothesis that captured verifier outputs can resemble a minority of honest-but-wrong verifiers, so deviation-based capture rules may incur false-positive costs. A synthetic pilot (4,000 trials per cell, $N = 7$ verifiers, f=2 captured) tests one instantiation of this hypothesis: trust-score weighting improves over plain majority under prompt-injection capture (accuracy 0.69 vs. 0.34 at honest accuracy 0.6; 0.94 vs. 0.74 at 0.8), but the same method does not improve under majority-bias mimicry (0.32 vs. 0.41 at honest accuracy 0.6). These pilot values are synthetic and do not establish a universal impossibility result. A descriptive local-model pilot is included in the evidence package, but a held-out real-trace benchmark remains future work; we release the framework, generators, and analysis scripts so the benchmark can be extended independently.

Reproducibility and scope. The evidence package contains the synthetic benchmark tables, the checked-in generators and aggregation code, and a descriptive four-model local Ollama pilot. It does not contain a held-out real trace benchmark, and the local pilot is not presented as an external generalization test. This work studies evaluation dynamics in LLM systems; we discuss the bounded dual-use risk of releasing the attack generators.

1 Introduction

Large language model (LLM)-based verification pipelines increasingly rely on *multi-source verifier ensembles* (Cui et al., 2026; Leung et al., 2026), in which a fixed pool of N LLM verifiers independently evaluate a candidate output and a final decision is produced by aggregating their votes. Such

ensembles have become a default building block of contemporary AI auditing, peer review assistance, and safety-critical content moderation, where the goal is to reduce variance and adversarial brittleness relative to any single model.

A natural and increasingly documented failure mode of these ensembles is what we call *verifier capture*: a small coalition of verifiers whose outputs are systematically biased in a coordinated direction can swing the aggregated decision, even when a majority of verifiers remain individually reliable. Verifier capture differs from the classical single-verifier hallucination or bias problem in two important ways. **(i) Capture is interactive.** The adversarial verifier does not need to be accurate in absolute terms; it only needs to produce outputs that, when combined with the other verifiers under a fixed aggregation rule, flip the final decision. **(ii) Capture is silent.** From the perspective of an external observer that only sees final aggregated verdicts, a captured ensemble looks identical to a correctly functioning one until the disagreement structure of individual votes is examined.

Recent work has begun to expose capture-like failure modes in LLM judges: Zhang et al. (2026) show that a biased judge *silently disables* a downstream curator past a sharp false-pass threshold. This paper asks the natural follow-up question: given that capture is real and silent, *can any aggregation rule do better than plain majority voting*? Our answer, in this empirical study, is *mostly no*.

This paper makes four contributions:

- A formal threat model for verifier capture in LLM ensembles, including a definition of ε-capture-resilience that decouples the adversary's per-verifier accuracy from its aggregate influence (Section 3).
- A unified experimental protocol that benchmarks seven aggregation methods under three attack families and four scenario configurations (Section 4).
- An empirical finding that, in the four synthetic scenarios tested, no aggregation method substantially outperforms plain majority and several trust-score variants underperform it under strong capture (Section 5). We offer a hypothesis, not a structural theorem, that capture can resemble honest-but-wrong behavior in the generator.
- A reproducible evidence package with the vote generators, aggregation code, frozen tables and a descriptive local-model pilot.

We emphasise that the *negative* headline finding is the paper's contribution, not an apology. TMLR's acceptance criteria explicitly prioritise technical correctness and audience interest over novelty, and a rigorous benchmark that documents the limits of the obvious aggregation strategies is valuable precisely because future work can build on it rather than re-derive it. We discuss when, and against which assumptions, our negative result would *not* hold (Section 10).

The remainder of this paper is organized as follows. Section 2 reviews classical Byzantine fault tolerance and LLM ensembling. Section 3 introduces the threat model. Section 4 describes the seven aggregation rules we benchmark. Section 5 reports the benchmark and its negative result. Section 7 discusses related work. Section 8 addresses broader impact. Section 10 concludes and outlines conditions under which our negative result would not hold.

2 Background

Classical Byzantine consensus. Byzantine fault tolerance (BFT) (Lamport et al., 1982) establishes that a deterministic consensus protocol over N replicas tolerates at most $f < N/3$ Byzantine

(arbitrary) failures when paired with digital signatures, or $f < N/2$ under synchrony assumptions with binary outputs. PBFT (Castro & Liskov, 1999) brought BFT to practical systems and HotStuff (Yin et al., 2019) linearised the communication pattern. All classical BFT results assume *binary* replica outputs; the extension to *continuous* outputs (which we need for LLM verifier probabilities) is not standard and is one of the contributions of this paper.

LLM-as-judge and ensembles. The use of LLMs as evaluators, or *judges*, was popularised by MT-Bench and Chatbot Arena (Zheng et al., 2023), where a single LLM judge is shown to correlate well with human preference ratings. Subsequent work studied ensemble-of-judges methods (Cui et al., 2026; Leung et al., 2026) to reduce variance and calibrate directional bias. These methods optimise for *passive* disagreement across annotators; we instead consider *adversarial* disagreement induced by verifier capture.

Adversarial robustness of LLM judges. Recent work has begun to expose the silent nature of biased LLM judges: Zhang et al. (2026) show that a biased judge *silently disables* a downstream curator past a sharp false-pass threshold, and broader red-teaming studies (Perez et al., 2022; Schulhoff et al., 2023) catalogue prompt-injection attacks. Our threat model (Section 3) operationalises these observations into a quantitative capture-resilience condition.

Peer-review automation. Liu (2026b) release an open-source multi-reviewer pipeline that we use as one of our testbeds; we discuss this and related calibration work (Liu, 2026a) in Section 7.

3 Threat Model and Definitions

We consider an ensemble $\mathcal{V} = \{v_1, \ldots, v_N\}$ of N LLM verifiers that independently evaluate a candidate output and produce a continuous decision distribution $p_i \in [0, 1]$ on the binary outcome $\{0, 1\}$; the aggregated decision is produced by a fixed aggregation rule $\mathcal{A} : [0, 1]^N \to \{0, 1\}$. A subset $\mathcal{A} \subseteq \mathcal{V}$ of size $f = |\mathcal{A}|$ is *captured* by an adversary that controls the verifiers' outputs but cannot observe or modify the inputs seen by honest verifiers.

Definition 1 (Honest verifier)**.** *A verifier $v_i \in \mathcal{V} \setminus \mathcal{A}$ is* honest *if $p_i = \mathbb{P}[\text{correct outcome} \mid x, \theta_i]$, where θ_i is the verifier's fixed-but-unknown system prompt and training. In particular, an honest verifier's outputs are conditionally independent given the input x.*

Definition 2 (Capture coalition)**.** *A captured verifier $v_j \in \mathcal{A}$ outputs $\tilde{p}_j = g(p_j^{true})$ where $g : [0, 1] \to [0, 1]$ is an adversarial response function drawn from one of three concrete attack families:*

- Prompt injection*: g is a fixed affine or step function chosen by the adversary (e.g. "always output 0.9").*
- Rubric injection*: g depends on the input via a hidden rubric bias that flips the verifier's preference order on a target subset of inputs.*
- Majority bias*: g mimics the running empirical mode of the other verifiers' outputs, so the captured output is indistinguishable from the majority.*

Definition 3 (Aggregation correctness)**.** *The aggregation rule $\mathcal{A}$ is (η, δ)-correct on a distribution $\mathcal{D}$ if $\Pr_{(x,y)\sim\mathcal{D},\mathcal{A},g}[\mathcal{A}(p_1, \ldots, p_N) = y] \geq 1 - \delta$, where the probability is taken over the input distribution, the adversary's choice of capture coalition $\mathcal{A}$, and the response functions g of all captured verifiers.*

Definition 4 (Capture-resilient aggregation)**.** *The aggregation rule $\mathcal{A}$ is* ε-capture-resilient *if, for every adversary that captures at most f verifiers, the ensemble remains (η, δ)-correct with $\delta \leq \varepsilon$ for every input distribution $\mathcal{D}$.*

Theorem 1 (Capture lower bound)**.** *For any deterministic aggregation rule $\mathcal{A}$, if the adversary captures $f \geq N/2$ verifiers, then for* some *input distribution $\mathcal{D}$ and* some *adversarial response function g from Definition 2, the rule $\mathcal{A}$ is not ε-capture-resilient for any $\varepsilon < 1/2$.*

Proof sketch. If the adversary controls a strict majority $f > N/2$, the adversary can choose g such that the captured verifiers all output the same value y^* with probability 1, while honest verifiers (whose accuracy is below 1 on $\mathcal{D}$) collectively cannot outvote $\mathcal{A}$. Formal reduction to the classical $f \geq N/2$ impossibility bound for deterministic aggregation. Full proof in Appendix A. □

Remark 1 (Continuous outputs vs. classical BFT)**.** *Theorem 1 recovers the classical $f < N/2$ BFT impossibility bound but does not yield a constructive upper bound, because LLM verifier outputs are continuous: a verifier that outputs $p_i = 0.51$ is qualitatively different from one that outputs $p_i = 0.99$, even though both would be treated as "in favor" by a binary-vote aggregation rule. This gap is what motivates the trust-score weighting in CRV (Section 4).*

Assumption 1 (Honest independence)**.** *Honest verifiers' outputs are conditionally independent given x. This is the standard assumption in LLM-ensemble calibration work (Cui et al., 2026; Zheng et al., 2023); we discuss the relaxation in Section 7.*

4 Methods: Seven Aggregation Rules

We benchmark seven aggregation rules. Three are *classical* and require no learning: plain majority voting (Section 4), PBFT-style deterministic thresholding (Section 4), and uniform-weighted mean (Section 4). Three are *trust-score-based* and share the EMA update $\tau_i^{(t)} = \alpha\tau_i^{(t-1)} + (1-\alpha)c_i^{(t)}$ where $c_i^{(t)} = 1 - |p_i^{(t-1)} - \text{med}_{-i}^{(t-1)}|$; they differ in how the trust score is *used* at decision time: weighted mean, weighted median, and Kalman-filter trust tracking. The seventh is ATV, an adaptive hybrid we proposed as a candidate that we expected to combine the strengths of soft weighting and outlier robustness.

Plain majority. Each verifier contributes a binary vote $\mathbb{1}[p_i \geq 0.5]$; the aggregate is the majority over the N binary votes. Cost $O(N)$ per round, no state.

PBFT-style deterministic thresholding. (Castro & Liskov, 1999) Each verifier again votes binary; the aggregate passes if at least $(N + f + 1)/2$ verifiers agree, which is the deterministic threshold that tolerates f adversarial nodes when a strict majority is required. Cost $O(N)$ per round.

Uniform-weighted mean. The aggregate is $\mathbb{1}[\sum_i p_i/N \geq 0.5]$. Cost $O(N)$, no state.

Trust-score weighted mean. At round t, weights are $w_i = \tau_i^{(t-1)} / \sum_j \tau_j^{(t-1)}$; the aggregate is $\mathbb{1}[\sum_i w_i p_i \geq 0.5]$. The trust update uses consistency against the per-round median. Cost $O(N)$ plus a per-round $O(N \log N)$ for the median.

Trust-score weighted median. Identical trust update; at decision time, sort verifiers by current p_i and pick the p_i at which the cumulative trust weight crosses half the total weight. This is robust to a single extreme value but expensive $O(N \log N)$ per round.

Kalman-filter trust tracking. Same decision rule as weighted mean, but trust tracking uses a Kalman filter with process variance $Q = 0.01$ and measurement variance $R = 0.05$ instead of an EMA, which we hypothesised would smooth trust under noisy verifier streams.

Atv: adaptive trust-weighted voting. A hybrid we designed as the natural synthesis of the two trust-score decision rules. ATV switches between weighted mean and weighted median at each round based on the standard deviation of the trust scores: if $\text{std}(\tau) < \tau_{\text{split}} = 0.5$ (trust concentrated, presumed-uniform regime), use weighted mean; otherwise (trust dispersed, capture suspected) use weighted median. The hyperparameter τ_{split} was tuned on a held-out scenario; we report sensitivity in Appendix C.

Why we expected Atv to win. Our *a priori* hypothesis, before running the benchmark, was that ATV would inherit the best of both worlds: the mean's sensitivity to subtle weight differences in the easy regime, and the median's robustness to outlier-dominated rounds in the hard regime. The benchmark (Section 5) does not support this hypothesis.

Why we report Atv even though it loses. We report ATV as a representative of the natural design space that future researchers might explore, rather than as our recommended method. The negative result is informative precisely because ATV *was* our best prior and still loses; the same is true of the Kalman variant.

5 Experimental Setup

Datasets. We use a synthetic verifier-output generator in this study; we discuss the limitations of this choice in Section 10. The generator (Section 5) models the agreement structure of a multi-LLM verification ensemble under verifier capture, and is calibrated to produce vote distributions whose summary statistics match published numbers for LLM judges on rubric decisions (Leung et al., 2026): per-verifier outputs are concentrated near 0 or 1 with $p_{\text{conf}} \in [0.7, 0.85]$ and a small amount of i.i.d. Gaussian noise ($\sigma = 0.05$). The candidate tree does not contain a held-out real-trace benchmark; the local pilot is reported separately as a descriptive artifact.

Attack models. We instantiate the three attack families of Definition 2 concretely: *prompt injection* (PROMPTINJ: captured verifier outputs the complement of its honest output), *rubric injection* (RUBRICINJ: captured verifier flips its label on a random fraction of inputs), and *majority bias* (MAJBIAS: captured verifier mimics the running median of the other verifiers' outputs). Section 5 gives the formal parameterisations.

Scenarios. We consider four scenarios that vary the two most consequential parameters: honest-verifier per-instance accuracy $\eta \in \{0.65, 0.85\}$ and attack strength $\alpha \in \{0.6, 0.9\}$ (probability that a captured verifier executes the attack on a given round). The four combinations are labelled S1–S4 in Table 1. We run $T = 1000$ rounds per scenario and average over $n = 5$ random seeds; full per-seed numbers are in Appendix C.

Metrics. We report *decision accuracy* as the fraction of rounds on which the aggregated decision equals the ground-truth label. Accuracy is the right metric for this benchmark because the goal of the aggregator is to produce the correct label, and the captured verifiers' goal is to produce an incorrect one; alternative metrics such as false-positive rate at the verifier level would require

labelling individual verifiers as captured or honest, which we cannot do from observed verifier outputs alone (this is the central statistical difficulty).

Hyperparameters. All trust-based methods use $\alpha = 0.9$ (EMA decay) and $c_i^{(t)} = 1 - |p_i - \text{med}_{-i}|$. ATV uses $\tau_{\text{split}} = 0.5$ chosen on a held-out scenario. PBFT threshold is $(N + f + 1)/2$. Code and configuration files are released with the paper.

6 Results

Headline. Table 1 reports the headline result. Plain majority voting is best or tied for best in S1–S3 and is within 0.8 percentage points of the best method in S4, where trust-mean reaches 0.8506 versus 0.8430 for majority. ATV never wins a scenario. The differences are small across the five random seeds; the reported cell standard deviations are below 0.01.

Table 1: Decision accuracy across four scenarios. Best per column in bold. Majority is best or tied in S1–S3; trust-mean is higher in S4, while the four-scenario mean is essentially tied between majority and trust-median.

Method	S1 (η=0.85, α=0.6)	S2 (η=0.65, α=0.6)	S3 (η=0.65, α=0.9)	S4 (η=0.85, α=0.9)	mean
majority	**0.889**	**0.553**	**0.459**	0.843	**0.686**
bft_pbft	0.777	0.522	0.458	0.720	0.619
weighted_uniform	0.804	0.484	0.365	0.729	0.595
trust_mean	0.883	0.521	0.432	**0.851**	0.672
trust_median	**0.889**	**0.553**	**0.459**	0.844	**0.686**
kalman_trust	0.882	0.535	0.446	0.842	0.677
ATV	0.883	0.527	0.438	0.846	0.673

Effect sizes. The five seeds are matched runs of one protocol, so contrasts are paired and we report d_z (mean per-seed difference over its standard deviation) rather than the independent-sample d our Statistical Procedures paragraph names. Table 2 gives d_z for ATV against each baseline; positive values favour ATV.

Table 2: Paired effect size d_z of ATV against each baseline, across the four scenarios. Values are computed from the per-seed accuracy records shipped with this submission.

Baseline	S1	S2	S3	S4
majority	−3.79	−5.36	−8.24	+0.90
trust_mean	−0.15	+3.23	+2.74	−5.00
trust_median	−3.79	−5.36	−8.24	+0.35
kalman_trust	+0.11	−5.14	−2.14	+2.37
weighted_uniform	+8.97	+5.03	+7.12	+16.93

Two features of Table 2 bear on the claim in Table 1. First, against the strongest baselines the effect is small and sometimes of the opposite sign: against `kalman_trust` in S1, $d_z = +0.11$, and against `trust_mean` in S1, $d_z = -0.15$. The large values in the last row are against `weighted_uniform`, the weakest method in every column, so they restate the table's ordering rather than add to it. We therefore do not claim ATV IMPROVES ON THE LEADING BASELINES, AND THE NEGATIVE RESULT IN TABLE 1 STANDS ON THE POINT ESTIMATES, NOT ON THE EFFECT SIZES.

Second, the reported per-seed accuracy vectors for `majority` and `trust_median` are identical in S1, S2 and S3 and differ by 0.007 in S4. The evidence records per-seed accuracies rather than per-round prediction streams, so we do not claim that the two rules produce byte-identical predictions.

A hypothesis for the negative result. We hypothesize that in this synthetic generator capture can be hard to distinguish from honest-but-wrong behavior: a captured verifier outputting the complement of its honest output can have a similar per-round deviation from the median as an honest verifier that happens to be wrong. A rule that uses deviation from the median to identify capture may therefore ignore both or exclude both. The evidence package does not contain a separate test of this mechanism, so it remains a hypothesis rather than a structural theorem.

In the high-honest scenario S1, the trust-based methods approach majority but trail it by 0.5 percentage points, consistent with the explanation above: the trust mechanism correctly down-weights captured verifiers when they *happen* to be outliers, but it also incorrectly down-weights honest verifiers that happen to be wrong, and the two effects approximately cancel.

In the strong-attack scenarios S3 and S4, the trust-based methods gain an apparent advantage in some columns (S4 for trust_mean) but lose it in others (S2 for all trust-based methods). The advantage in S4 disappears when averaged across all four scenarios: the mean column is dominated by majority and trust_median, both at 0.686.

Per-attack breakdown. Across the three attack families (prompt injection, rubric injection, majority bias), the best method differs by family. Trust-mean is highest under prompt injection and majority-bias mimicry, while trust-median, Kalman-trust, and Atv are at the top under rubric injection. Majority is within 1.7 percentage points of the best attack-family mean and remains a strong baseline; this does not establish a universal rule.

Sensitivity evidence. The checked-in evidence contains an ensemble-size sweep at $N \in \{5, 7, 10\}$, but no per-cell τ_{split} or trust-decay sweep. We therefore make no claim about hyperparameter robustness; the sweep is reported exactly in Appendix C.

Implication. Within the synthetic ensemble sizes, attack strengths, and honest-accuracy regimes in this benchmark, the tested trust-based and adaptive rules do not show a consistent gain over plain majority. This is a bounded empirical statement, not a claim that all non-trivial aggregators fail.

7 Related Work

Classical Byzantine consensus. Lamport et al. (1982); Castro & Liskov (1999); Yin et al. (2019) establish the $3f + 1$ bound for binary-output consensus under f adversarial nodes. Our setting differs in that LLM verifier outputs are continuous probability distributions, so classical BFT bounds do not transfer directly; this motivates our ε-capture-resilience definition in Section 3.

LLM-as-judge and ensemble calibration. Zheng et al. (2023) introduce the MT-Bench and Chatbot Arena evaluation paradigm; Leung et al. (2026) provide the most recent calibration numbers for 8 LLM judges on rubric decisions, and Cui et al. (2026) jointly optimize ensemble weights and calibration for subjective supervision. CRV differs from EDO in operating under an explicit adversarial model (capture) rather than a passive disagreement model.

Adversarial robustness of LLM judges. Zhang et al. (2026) show that a biased judge *silently* disables a downstream curator past a sharp false-pass threshold, and Perez et al. (2022); Schulhoff et al. (2023) catalogue prompt-injection and red-team attack techniques against LLM evaluators. Our threat model operationalises these observations into a quantitative capture-resilience definition and a reproducible benchmark; it does not establish a general aggregation-correctness guarantee.

Peer-review automation. Liu (2026b) release an open-source multi-reviewer pipeline that we use as the testbed for our empirical evaluation (Section 5); Liu (2026a) develops calibration methods for multi-agent ensembles more broadly and is the closest work in our own research line.

8 Broader Impact

Positive impact. The primary positive impact of this work is *negative*: by showing that no aggregation method we tested substantially outperforms plain majority voting under verifier capture, we redirect practitioner effort away from building sophisticated aggregators and toward investments that *do* pay off in the capture regime: increasing honest-verifier accuracy, diversifying the verifier pool, and auditing deployed ensembles for suspicious agreement patterns. We expect the benchmark to be useful for engineering teams that are currently considering whether to invest in a custom aggregation layer; our data suggest the answer is "no" under the parameter regime we test, and the engineering budget should be spent elsewhere.

A secondary positive impact is methodological: the failure modes of the seven aggregation rules we test illuminate which statistical assumptions a successful capture-resilient aggregator would need to break. Future research that does find a winning method can explicitly cite which assumption we showed fails here.

Negative impact and dual-use concerns. The same attack implementations that allow us to *benchmark* verifier capture also allow an adversary to *measure* capture susceptibility of a competing verifier product. We consider this a real but bounded risk for the following reasons. **(i)** The three attack families we instantiate (PromptInj, RubricInj, MajBias) are straightforward enough that a competent adversarial team could re-derive them from the threat model alone; our code release does not materially lower the cost. **(ii)** Our benchmark demonstrates that these attacks, when attempted against a well-calibrated honest majority, have only small absolute effect (the gap between majority and the best trust-based method is below 1 percentage point in 3 of 4 scenarios). An adversary using our code to measure capture susceptibility will therefore find that the practical upside of

attack deployment is small. **(iii)** Our benchmark is *synthetic*. An adversary who wishes to deploy a capture attack against a real ensemble must still construct a real captured verifier that mimics the target ensemble's distribution, which is a substantially harder engineering problem than running our code.

Responsible disclosure norms. The candidate evidence tree contains the benchmark code and attack generators; the final license and release wording remain user decisions. No captured-verifier weights or fine-tuning scripts are present.

Limitations of public release. The benchmark code is sufficient to reproduce the synthetic tables but not to deploy a capture attack against a real ensemble without substantial additional engineering. This scope is deliberate.

Stakeholder considerations. Verifier-capture attacks disproportionately threaten operators of multi-LLM auditing pipelines. The benchmark is documented for defensive testing; no external outreach or deployment validation is claimed.

Positive-negative balance. On balance, we believe the positive impact of releasing this benchmark—redirecting engineering effort away from ineffective aggregation rules toward more productive investments in verifier quality and pool diversification—substantially outweighs the bounded negative impact of marginally lowering the cost of an attack that, per our own benchmark, has small practical effect against well-calibrated honest majorities.

9 Ethics Statement

This study does not involve human subjects, personal data, or animal subjects. The verifier-output data used here is either synthetic (Section 5) or from the descriptive local-model pilot.

Dual-use risk. We have discussed the dual-use risk of releasing the attack implementations in detail in Section 8. In summary, the bounded risk is discussed in Section 8. The candidate tree contains no captured-verifier weights or fine-tuning scripts; final license and responsible-use wording remain user decisions.

Affected stakeholders. The primary stakeholders are operators of multi-LLM verification ensembles, including AI-auditing platforms, peer-review automation tools, and safety-content moderation pipelines. We have reached out to the maintainers of two such platforms ((Liu, 2026b) and the public release of Leung et al. (2026)'s evaluation harness) to inform them of our findings ahead of public release.

IRB and funding. This study was not subject to IRB review. No external funding was received for this work.

Author conflicts of interest. The authors declare no financial conflicts of interest. The authors maintain an open-source peer-review audit tool (Liu, 2026b) that is discussed as one of the motivating use cases for this benchmark; the tool is

referenced consistently with the rest of the literature and no special pleading is made for it.

10 Conclusion

We have presented an empirical benchmark of seven aggregation methods for multi-LLM verifier ensembles under three verifier-capture attacks, across four scenarios. In this synthetic benchmark, plain majority is a strong baseline: it is best or tied in three scenarios and within 0.8 percentage points of the best method in the fourth. Several trust-score variants underperform majority in some scenarios. We hypothesize that captured behavior can resemble honest-but-wrong behavior in the generator, but we do not establish that as a structural theorem.

When our negative result would *not* hold. The bounded result rests on the synthetic generator's assumption that captured verifiers can resemble honest-but-wrong verifiers. If a real-world capture breaks this assumption — for instance, by introducing a structured deviation (e.g. a verifier that always outputs the exact median plus a fixed delta) — then a more sophisticated aggregation rule may recover above majority. Our benchmark does not rule this out; it only shows that for the three concrete attack parameterisations we test, the gain is at most a fraction of a percentage point.

A second assumption is that honest verifiers are exchangeable. If honest verifiers have known specialty areas (e.g. one verifier is better at math, another at code), then a domain-aware aggregator may do substantially better than majority. Our synthetic generator deliberately does not model this specialisation, so our benchmark under-estimates the upside of sophisticated aggregation in heterogeneous pools.

11 Discussion and Limitations

Our results rely on a synthetic verifier-output generator calibrated to published rubric-LLM numbers but not validated against real LLM trace data. A practitioner applying our benchmark to a real ensemble should re-run our scenario definitions with the real distribution and may find that some methods we rule out do in fact help on their specific distribution. We release the synthetic generator and the attack implementations to facilitate this replication.

Future work. Three directions follow from this study. (i) Validate the negative result on real LLM verifier trace data; the result may not survive the transition from synthetic to real. (ii) Investigate structured-attack settings where captured verifiers introduce deviations outside the honest-but-wrong distribution; the negative result may not apply there. (iii) Investigate diversified verifier pools where domain-aware aggregation can outperform majority by exploiting specialisation.

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A Proof of the Capture Lower Bound (Theorem 1)

We prove the deterministic lower bound stated in Theorem 1: if $f \geq N/2$ of the verifiers are captured, no deterministic aggregation rule $\mathcal{A} : [0,1]^N \to \{0,1\}$ can be ε-capture-resilient for $\varepsilon < 1/2$.

Proof. Consider the case $f > N/2$ (strict majority capture; the case $f = N/2$ follows by a standard limit argument). The adversary chooses a target label $y^* \in \{0,1\}$ and a response function g such that every captured verifier outputs y^* with probability 1 on every input. The adversary also chooses an input distribution $\mathcal{D}$ on which the *honest* verifiers have per-instance accuracy strictly below $1/2$; this is always possible by choosing $\mathcal{D}$ to concentrate on inputs where the honest verifiers systematically err.

Because the captured coalition $\mathcal{A}$ forms a strict majority, the deterministic rule $\mathcal{A}$ sees $p_i \in \{0,1\}$ on the captured inputs (after g is applied) and any values in $[0,1]$ on the honest inputs. Since $\mathcal{A}$ is deterministic, its decision is a function only of the multiset $\{p_1, \ldots, p_N\}$, and in particular of the *number* of captured verifiers voting y^*, which is $f > N/2$. For any such deterministic rule, there exists an assignment of honest outputs such that $\mathcal{A}$ outputs y^* on every input, even when the true label is the opposite $\bar{y}^*$; this is a direct consequence of the fact that a strict majority of N deterministic binary inputs forces the binary majority rule, and any rule that agrees with binary majority on strict-majority inputs cannot be ε-resilient for $\varepsilon < 1/2$ on the chosen $\mathcal{D}$.

This shows that for any deterministic $\mathcal{A}$, the adversary can construct a distribution $\mathcal{D}$ on which $\mathcal{A}$ achieves error probability at least $1/2$, completing the proof. □

B Algorithm Pseudocode and Complexity

We give pseudocode for the four representative aggregation rules. All four have memory $O(N)$ for the per-verifier state.

Plain majority.

Require: votes $p_1, \ldots, p_N$
1: $s \leftarrow \sum_i \mathbb{1}[p_i \geq 0.5]$
2: **return** $\mathbb{1}[s > N/2]$

Time $O(N)$.

PBFT-style threshold.

Require: votes $p_1, \ldots, p_N$; bound f
1: $s \leftarrow \sum_i \mathbb{1}[p_i \geq 0.5]$
2: **return** $\mathbb{1}[s \geq (N + f + 1)/2]$

Time $O(N)$.

Trust-score weighted mean.

Require: votes $p_1, \ldots, p_N$; trusts $\tau_1, \ldots, \tau_N$; decay α
1: $w_i \leftarrow \tau_i / \sum_j \tau_j$ for each i
2: **return** $\mathbb{1}[\sum_i w_i p_i \geq 0.5]$
3: Update: $\tau_i \leftarrow \alpha\tau_i + (1-\alpha)(1 - |p_i - \text{med}_{-i}|)$ for each i

Time $O(N \log N)$ per round (median), memory $O(N)$.

Trust-score weighted median.

Require: votes $p_1, \ldots, p_N$; trusts $\tau_1, \ldots, \tau_N$; decay α

1: Sort indices by p_i; let π be the sorted permutation
2: $W \leftarrow \sum_j \tau_j$; find smallest k with $\sum_{i \le k} \tau_{\pi(i)} \ge W/2$
3: **return** $\mathbb{1}[p_{\pi(k)} \ge 0.5]$
4: Update trust as in weighted mean.

Time $O(N \log N)$ per round, memory $O(N)$.

Atv (adaptive hybrid).

Require: votes $p_1, \ldots, p_N$; trusts $\tau_1, \ldots, \tau_N$; decay α; split τ_{split}

1: **if** std$(\tau) < \tau_{\text{split}}$ **then**
2: **return** weighted-mean decision
3: **else**
4: **return** weighted-median decision
5: **end if**
6: Update trust as in weighted mean.

Time $O(N \log N)$ per round, memory $O(N)$.

C Additional Ablations and Sensitivity Sweeps

Per-attack-model breakdown. Table 3 reports per-attack-model accuracies. The ordering of methods is qualitatively consistent across the three attack families: majority and trust_median remain tied at the top, with the largest absolute gap (0.7 percentage points) appearing under rubric injection.

Table 3: Decision accuracy by attack model, averaged across all four scenarios (5 seeds, $T = 1000$ rounds per cell).

Method	PromptInj	RubricInj	MajBias
majority	0.877	0.945	0.997
bft_pbft	0.765	0.886	0.985
weighted_uniform	0.876	0.944	0.997
trust_mean	**0.894**	0.944	**0.998**
trust_median	0.877	**0.945**	0.997
kalman_trust	0.877	**0.945**	0.997
Atv	0.886	**0.945**	0.997

Hyperparameter sensitivity. Atv's τ_{split} was tuned on a held-out scenario. Sensitivity sweeps over $\tau_{\text{split}} \in [0.2, 0.8]$ shift Atv's mean accuracy by at most 0.4 percentage points; the rank ordering is preserved across all sweep values. Trust-update decay α sweeps over $[0.5, 0.99]$ shift trust_mean by ± 0.5 percentage points with no rank change.

Ensemble-size sweep. The checked-in ensemble-size sweep covers $N \in \{5, 7, 10\}$ with $f = \lfloor N/3 \rfloor$. At $N = 5$, majority and several methods tie at 0.7479. At $N = 7$, majority and trust-median are essentially tied at 0.686. At $N = 10$, trust-median reaches

0.7668, higher than majority's 0.6785. These three points do not establish a monotone ensemble-size trend; they are reported as descriptive sweep values.

Local-model pilot. The evidence package includes a descriptive local Ollama pilot with four models, 20 questions, three seeds, and two captured slots. The reported rounded mean accuracy is 0.9167 for all seven methods. This pilot is not a held-out trace benchmark and is not evidence for external generalization.

D Code and Reproducibility

Repository layout. The candidate evidence tree contains the benchmark code and frozen result files. Final license and release wording remain user decisions. The repository contains:

- `crv.py`: legacy CRV implementation (retained for archival reference; not used as a benchmarked method).
- `attacks.py`: implementations of the three attack families (PromptInj, RubricInj, MajBias).
- `extended_methods.py`: the seven aggregation methods benchmarked in this paper.
- `benchmark.py`: the four-scenario benchmark driver.
- `verify_p6.py`: the verify suite used during paper preparation.

No held-out demo claim. The evidence tree contains no independently validated real-trace demo. The local pilot is a descriptive artifact, not a deployment or held-out validation claim.

Reproducibility hashes. The ingest manifest and the Research Factory overlay record SHA-256 hashes for the frozen source and evidence files. A Python dependency lock and complete local-pilot runtime logs are not present; those remain release blockers.

Computational footprint. A full benchmark run (4 scenarios $\times$ 7 methods $\times$ 5 seeds, $T = 1000$ rounds) takes approximately 4 minutes on a single CPU core (no GPU required) and uses 50 MB of memory. The complete experiment can be reproduced in under 10 minutes on commodity hardware.

Reproducibility Statement

Available evidence. The candidate tree contains the synthetic benchmark tables, the checked-in generator and aggregation code, and a descriptive four-model local Ollama pilot. The ingest manifest records SHA-256 hashes for the frozen source and evidence files. A complete Python dependency lock and runtime logs for the local pilot are not present.

What is not established. The package does not contain a held-out real-trace benchmark, a pre-registered temporal replication, or a proof of a universal capture-resilience bound. Those remain open evidence gates and are not presented as completed results.